Kiwi Mainboard Datasheet (Work In Progress)

1. Features

- Raspberry Pi RP2350B Microcontroller with 2 to 4 MB program memory
- 5V Supply Voltage, 5V tolerant I/O[1]
- Low power operation (< 1W)
- 60 mm × 60 mm single board
- Temperature Range: −40°C to 150°C
- Power over USB or stack connector
- Connectivity:
  - IEEE 802.11n 2.4GHz WiFi
  - Bluetooth 5.2 with BLE
  - Integrated CAN transceiver
- Micro-SD storage up to 16GB
- Accelerometer, Gyroscope, Magnetometer, Humidity, Air Quality, Temperature, Pressure, Flux sensors
- Stackable for further I/O expansion
- Supports ARM Serial Wire Debug, compatible with Raspberry Pi Debug Probe

2. Applications

Kiwi Mainboard is a highly integrated module geared towards

- Education
- Table-top experiments
- Environmental sensing
- Sub-orbital experiments

3. Description

Kiwi Mainboard is a compact, low-power microcontroller board providing core functionality to the Kiwi family of laboratory, bench-top flat satellites; and environmental sensing and communication to the RAIL family of sub-orbital experiments.

The board includes a variety of sensors, modes of communication, and exposes many general purpose input/output (GPIO) pins for expansion. It is designed to be stackable, allowing expansion of capabilities through additional boards.

[1] The GPIO pins are 5V tolerant (up to 5.5V) only when `IOVDD` is present and is 3V3. The GPIO pins are only 3V3 failsafe otherwise. Note, the on-board 5V to 3V3 regulator takes time to power up and stabilize and `IOVDD` is absent at this time. Refer to the Raspberry Pi RP2350B Datasheet for more details.

4. Contents

5. Revisions

Rev. B1 - 2025-10-06

- Hardware
  - Replaced Bosch BMP390 with Measurement Specialty MS560702BA03-50 for pressure and altitude measurement.
- Documentation
  - First release.

Rev. B0 - 2025-09-15

- Hardware
  - Integrated microcontroller instead of using separate stamp.
  - Added micro SD card slot.

Rev. A0 - 2025-07-30

- Hardware
  - Internal hardware release using the Chickadee Stamp.
  - Flight qualified on PICTURE-D mission as CHARIOT.
- Software
  - Memory safe, event-driven, asynchronous firmware written in Rust.

6. Specifications

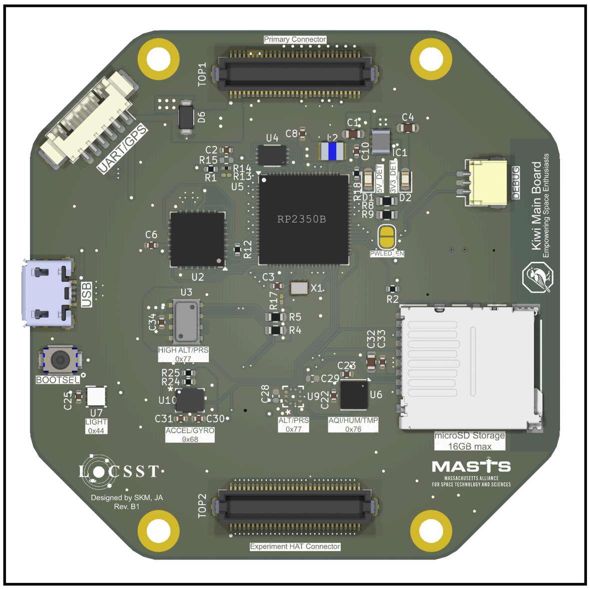

Figure 1: Top View

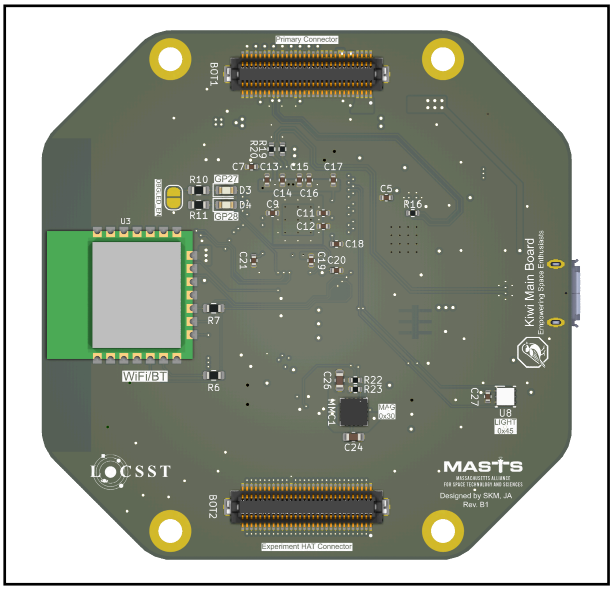

Figure 2: Bottom View

6.1. Specifications

Table 1: Physical Characteristics

Parameters	Minimum	Typical	Maximum	Unit	Notes
Width	—	60	—	mm	—
Height	—	60	—	mm	—
Weight	—	TBD	—	g	—
Stand-off Height Above	—	5	—	mm	—
Stand-off Height Below	5	5	6	mm	Increments of 0.5 mm

Table 2: Electrical Specifications

Parameters	Symbol	Minimum	Typical	Maximum	Unit	Condition
Operating Voltage	V_{IN}	4	5	5.5	V	—
Operating Current	I	—	250	—	mA	Power over USB, sensors running at 50Hz cadence with TCP server serving data over WiFi at 22°C

6.2. Absolute Maximum Ratings

Table 3: Absolute Maximum Ratings

Parameter	Symbol	Minimum Value	Maximum Value	Unit	Note
Power Supply Voltage	V_{IN}	−0.3	6.2	V	
Ambient Temperature	T_A	−40	85	°C	

7. Detailed Description

7.1. Overview

Kiwi Mainboard is a System-on-a-Module (SoM) around the Raspberry Pi RP2350B microcontroller. Unlike a traditional microcontroller breakout board that provides access to the various I/O pins on the microcontroller, the SoM includes common environmental sensors and communication capabilities. This makes the SoM suitable for table-top experiments without the need of additional hardware.

The SoM includes an accelerometer, a gyroscope, a magnetometer, an air quality and humidity sensor, a pressure and temperature sensor that also functions as an altimeter, and two light flux sensors on either sides of the module. The SoM has a micro-SD card slot, allowing it to store data up to 16 GiB. The integrated Raspberry Pi RM2 provides wireless connectivity through 2.4GHz, single channel WiFi or Bluetooth 5.2 with Low Energy.

The SoM can be powered via a USB micro-B port connected to a wall-wart or a PC. Firmware can be uploaded to the SoM over USB. The SoM can provide one or more UART interfaces over USB for communication depending on firmware configuration. The SoM leverages the commercially available Raspberry Pi Debug Probe for advanced programming and debugging capabilities offered by the Arm Serial-Wire-Debug (SWD) protocol.

The SoM is compatible with the Chickadee Stackable Bus system. The stack connectors can provide power and communication over CAN protocol, which allows multiple SoMs on the stack to communicate with each other. 28 `GPIO` pins are accessible on the stack connector for the microcontroller to connect to additional sensors and peripherals on the stack and expand its capabilities.

7.2. Functional Block Diagram

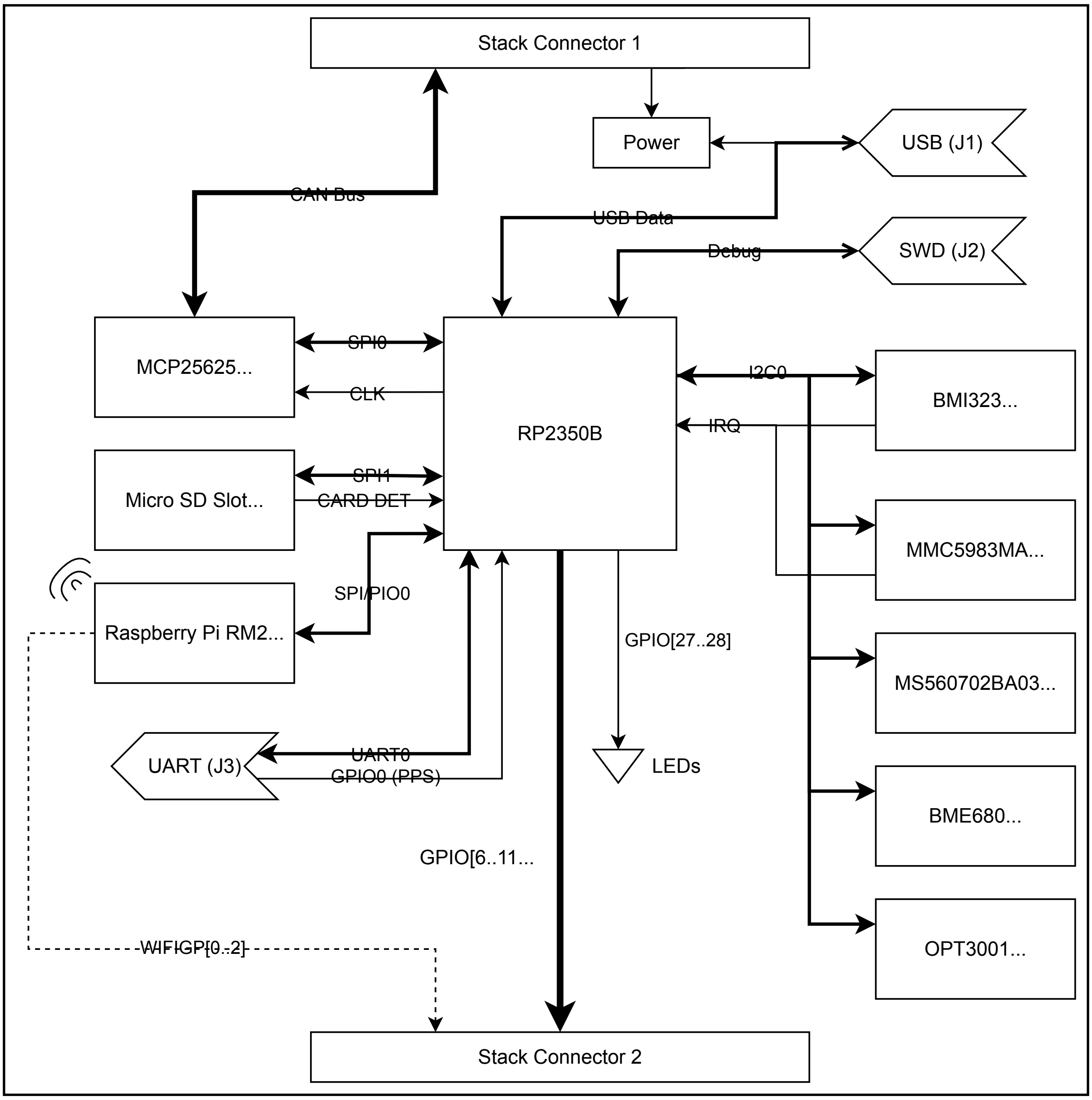

Figure 3: Block diagram of Kiwi Mainboard.

7.3. Interfaces and Pin Configuration

User servicable connectors are located on the top surface of the board (see Figure 1).

7.3.1. USB (J1)

USB-micro B connector for power and USB data. Supports USB 1.1 Full Speed (12 Mbps) device mode. The USB port is used for uploading UF2 firmware when the board is booted into flash storage mode by holding down the `BOOTSEL` button before powering up the board. The USB port also exposes multiple COM/UART devices to a connected host device via the USB CDC-ACM mode, if configured as such in the device firmware.

7.3.2. Arm Serial Wire Debug (J2)

1 mm pitch, 3-pin JST SH connector for ARM Serial Wire Debug (SWD) interface. Compatible with Raspberry Pi Debug Probe. Connect to the Debug Probe "D" port using a 3-pin JST SH cable available with the probe. Refer to the Debug Probe documentation for the pinout of the port.

7.3.3. UART Connector (J3)

1.25 mm pitch Molex® PicoBlade 53261-0671 6-pin, horizontal connector to provide a UART link (logic level 3V3) without hardware flow control. 5V (unregulated) and 3V3 (regulated) power is also available on this connector for the downstream device.

Table 4: Pinout of UART Connector (J3)

Pin	Function	Description
1	+5V	Unregulated/externally regulated bus voltage
2	+3V3	3.3V power, 300 mA max.
3	RX	Receive data from UART device
4	TX	Send data to UART device
5	PPS	Connected to `GPIO0` of the RP2350B. Intended for use as PPS input from a GPS device.
6	GND	Ground return pin.

7.3.4. Bottom Stack Connector 1 (`BOT1`)

Amphenol Bergstack 10132798-062100LF 60-pin connector that mates with the board below it, allowing stack heights of 5 mm, 5.5 mm or 6 mm (depending on the height of the complimentary `TOP` connector on the board underneath it in the stack). This connector provides power to the SoM, and continues the CAN bus signals downstream.

Table 5: Pinout of Stack Bottom 1 Connector (BOT1)

Pin	Function	Description
1–16	+5V	Externally regulated power input
18, 21, 24, 28–30, 32, 37, 43, 49–52, 57–60	GND	Ground return for power
22	CAN+	Positive wire of the CAN differential pair
23	CAN-	Negative wire of the CAN differential pair
17, 19–20, 25–27, 31, 33–36, 38–39, 41–42, 44–48, 53–56	Reserved	Pass from `BOT1` to `TOP1`, DO NOT CONNECT elsewhere.

7.3.5. Top Stack Connector 1 (`TOP1`)

Amphenol 10132797-065100LF 60-pin connector that allows a gap (stack height) of 5 mm with the board stacked above it. This connector passes through power for the stack upwards (received through `BOT1`), and continues the CAN bus signals downstream. The pinout of this connector is identical to `BOT1`.

7.3.6. Bottom Stack Connector 2 (`BOT2`)

Amphenol Bergstack 10132798-062100LF 60-pin connector that mates with the board below it, allowing stack heights of 5 mm, 5.5 mm or 6 mm (depending on the height of the complimentary `TOP` connector on the board underneath it in the stack). This connector breaks out available GPIO pins from the SoM.

Table 6: Pinout of Stack Bottom 2 Connector (BOT2)

Pin	Function	Description
33	`WIFI_GP2`	GPIO pins from the RP2350B microcontroller. The number corresponds to the GPIO number (not the physical pin number). `WIFI_GP#` are GPIO pins from the RM2 module, and can not be directly accessed by the microcontroller.
34	`WIFI_GP1`	
35	`WIFI_GP0`	
36	`GPIO47`	
37	`GPIO46`	
38	`GPIO45`	
39	`GPIO44`	
40	`GPIO43`	

Pin	Function	Description
41	GPIO42	
43	GPIO36	
44	GPIO37	
45	GPIO38	
46	GPIO39	
47	GPIO40	
48	GPIO41	
49	GPIO22	
50	GPIO15	
51	GPIO14	
52	GPIO13	
53	GPIO12	
54	GPIO11	
55	GPIO10	
56	GPIO9	
57	GPIO8	
58	GPIO7	
59	GPIO6	
1, 30, 31, 42, 60	GND	Signal ground return.
2–29, 32	Reserved	Pass from `BOT2` to `TOP2`, DO NOT CONNECT elsewhere.

7.3.7. Top Stack Connector 2 (`TOP2`)

Amphenol 10132797-065100LF 60-pin connector that allows a gap (stack height) of 5 mm with the board stacked above it. The GPIO pins from the SoM are available on this conector. The pinout of this connector is identical to `BOT2`.

7.3.8. I^{2}C Address Table

The various sensors present on the SoM are addressable over the I^{2}C bus (RP2350B peripheral `I2C0`). Each sensor has a unique address that is used to communicate with it. The addresses of various sensosrs are tabulated in Table 7.

Sensor	Function	Address
Bosch BMI323	Accelerometer, Gyroscope	`0x68`
Memsic MMC5983MA	Magnetometer	`0x30`
TE MS560702BA03-50	Precision barometer	`0x77`
Bosch BME680	AQI, Humidity, Temperature sensor	`0x76`
TI OPT3001	Light flux sensor, SoM top	`0x44`
TI OPT3001	Light flux sensor, SoM bottom	`0x45`

7.4. Capabilities

7.4.1. Microcontroller

The microcontroller features two Arm Cortex M33 or two Hazard3 RISC-V CPU cores with hardware, IEEE-754 single-precision (32-bit) floating point math support, running at 150 MHz, with 520 KiB of on-chip static random access memory (SRAM). The microcontroller offers 2×UART, 2×SPI and 2 ×I²C hardware peripherals, along with 24×PWM channels and 8×ADC channels. Additionally, the RP2350B offers 3×PIO blocks, each with 4 PIO state machines. The PIO subsystem is programmable to extend the I/O capabilities of the RP2350B beyond the hardware peripherals already present on the chip. For further details on the Raspberry Pi RP2350B, refer to its datasheet.

7.4.2. Inertial Measurement Unit (IMU)

The SoM includes a Bosch BMI323 inertial measurement unit. An inertial measurement unit measures the inertial properties of itself, in this case the 3-axis (Cartesian) acceleration and angular speed. The relevant performance metrics of the IMU is provided in Table 8. The IMU is connected to the microcontroller over the I²C bus (`I2C0`), at address `0x68`.

Table 8: Capabilities of the Bosch BMI323 IMU

Parameters	Condition	Min.	Typical	Max.	Unit
Power on time	Time from supply "on" to serial I/F operational		1.5		ms
Operating Temperature		−40		+85	°C
Accuracy of output data rate	Any mode with gyroscope enabled in any state			1.7	%
	Acceleration only, high performance mode			2	%
	Combo mode, high performance		0.0037		$\frac{\%}{°C}$
Accelerometer					
Start-up time	Suspend to high performance mode		2		ms
Range	Selectable via serial command		±2		g
			±4		
			±8		

Parameters	**Condition**	**Min.**	**Typical**	**Max.**	**Unit**
			±16		
Resolution			16		bit
Sensitivity	Range: $\pm 2g$		16384		$\frac{\text{LSB}}{g}$
	Range: $\pm 4g$		8192		
	Range: $\pm 8g$		4096		
	Range: $\pm 16g$		2048		
Sensitivity error	Soldered, over lifetime		± 0.5		%
Zero-g offset	soldered		± 35		mg
	soldered, over lifetime		± 50		
Noise density	High performance mode, range $\pm 8g$		180		$\frac{\mu g}{\sqrt{\text{Hz}}}$
Output Data Rate	High performance and normal mode	12.5		6400	Hz
	High performance and normal mode	0.78125		400	
Cross-axis sensitivity	Non-orthogonality among axes, evaluated as lower triangular matrix		± 0.3		%
	Alignment error, relative to package outline		± 0.5		°
Gyroscope					
Start-up time	Suspend to high performance mode		30		ms
Range	Selectable via serial command		±125		$° / s$
			±250		
			±500		
			±1000		
			±2000		
Resolution			16		bit
Sensitivity	Range: $\pm 2000°/s$		16.384		$\frac{\text{LSB}}{° / s}$
	Range: $\pm 1000°/s$		32.678		
	Range: $\pm 500°/s$		65.536		
	Range: $\pm 250°/s$		131.072		
	Range: $\pm 125°/s$		262.144		
Sensitivity error	Soldered, over lifetime, after self-calibration		± 0.7		%
	Soldered, over lifetime, without self-calibration		± 3		
Zero-rate offset	soldered, over lifetime		± 1		$° / s$

Parameters	Condition	Min.	Typical	Max.	Unit
Noise density	High performance mode		0.007		$\frac{° / s}{\sqrt{\mathrm{Hz}}}$
Output Data Rate	High performance and normal mode	12.5		6400	Hz
	High performance and normal mode	0.78125		400	
Cross-axis sensitivity	Non-orthogonality among axes, evaluated as lower triangular matrix		± 0.3		%
	Alignment error, relative to package outline		± 0.5		°
Zero rate offset error, gravity induced	Gravitation $(1g)$ parallel to each main axis			0.1	$° / s$
Temperature					
Resolution			16		bits
Range		−41		87	°C
Output at 23°C			0		LSB
Sensitivity			512		$\frac{\mathrm{LSB}}{\mathrm{K}}$
Temperature offset			± 3	± 4	K
Temperature sensitivity error			2	4.5	%
Output Data Rate	Any power operation mode with gyroscope in high performance or normal mode			50	Hz
	Any power operation mode with gyroscope either disabled, in low-power mode or drive-only mode			12.5	
	Accelerometer in low power mode, gyroscope disabled			6.25	

7.4.3. Magnetometer

The SoM includes a Memsic MMC5983MA magnetometer. A magnetometer measures the magnetic field around itself in 3-axis (Cartesian). The relevant performance metrics of the magnetometer is provided in Table 9.

Table 9: Capabilities of the Memsic MMC5983MA Magnetometer

Parameters	Condition	Min.	Typical	Max.	Unit
Power on time			5		ms
Operating Temperature		−40		+105	°C
Range			± 8		Gauss
Resolution			18		bit

Parameters	Condition	Min.	Typical	Max.	Unit
Total RMS noise	BW = 00		0.4		mG
	BW = 00		0.6		
	BW = 00		0.8		
	BW = 00		1.2		
Output Data Rate	BW = 00		50		Hz
	BW = 01		100		
	BW = 10		225		
	BW = 11		580		
	BW = 11, CM Freq. = 111		1000		
Heading accuracy			± 1		°
Sensitivity	16-bit		4096		$\frac{\mathrm{LSB}}{\mathrm{G}}$
	18-bit		16384		
Sensitivity accuracy	Range: ± 8G		± 5		%
Alignment error			± 1	± 3	°
Maximum exposed field			10000		G

7.4.4. Pressure and Altitude Sensor

The TE Connectivity MS560702BA03-50 is a precision barometer capable of measuring pressure as low as 10 hPa (10 mbar). Using a precision temperature measurement, altitude can be derived from the pressure measurement. The sensor is connected to the microcontroller over the I^{2}C bus (`I2C0`), at address `0x77`.

Table 10: Capabilities of the TE MS560702BA03-50 Pressure and Temperature Sensor

Parameters	Condition	Min.	Typical	Max.	Unit
Operating temperature		−40		+85	°C
Range	Full accuracy	300		1100	mbar
	Extended, linear range of ADC	10		2000	
Resolution			24		bit
Conversion time	OSR = 4096	7.40	8.22	9.04	ms
	OSR = 2048	3.72	4.13	4.54	
	OSR = 1024	1.88	2.08	2.28	
	OSR = 512	0.95	1.06	1.17	
	OSR = 256	0.48	0.54	0.60	
Pressure Sensitivity	OSR = 4096		0.024		mbar
	OSR = 2048		0.036		
	OSR = 1024		0.054		
	OSR = 512		0.084		
	OSR = 256		0.130		

Parameters	Condition	Min.	Typical	Max.	Unit
Temperature Sensitivity	OSR = 4096		0.002		°C
	OSR = 2048		0.003		
	OSR = 1024		0.005		
	OSR = 512		0.008		
	OSR = 256		0.012		

7.4.5. Air Quality, Humidity, and Temperature Sensor

Bosch BME680 is an integrated environmental sensor that combines gas, humidity, pressure and temperature sensing. The sensor is connected to the microcontroller over the I^2C bus (`I2C0`), at address `0x76`. The relevant performance metrics of the sensor is provided in Table 11.

Parameters	Condition	Min.	Typical	Max.	Unit
Start-up time	Time to first communication after both V_{DD} > 1.58 V and V_{DDIO} > 0.65 V		2		ms
Gas Sensor					
Operating Range	Temperature	−40		85	°C
	Humidity	10		95	%RH
Response time	Ultra-low power mode		92		s
	Low power mode		1.4		
	Continuous mode		0.75		
Noise			1.5		%
Air Quality Sensor					
Range		0		500	
Humidity Sensor					
Measurement Range	Temperature	−40	25	85	°C
	Humidity	0		100	%RH
Full Accuracy Range	Temperature	0		65	°C
	Humidity	10		90	%RH
Resolution			0.008		%RH
Noise	Highest oversampling		0.01		%RH
Pressure Sensor					
Measurement Range		300		1100	hPa
Operating Temperature	Operational	−40	25	85	°C
	Full accuracy	0		65	
Resolution			0.18		Pa
Temperature Sensor					
Measurement Range		−40		85	°C
Absolute Accuracy	At 25°C		± 0.5		°C

Parameters	Condition	Min.	Typical	Max.	Unit
	0 – 65°C		± 1.0		
Resolution			0.01		°C
Noise	Lowest oversampling		0.005		°C

7.4.6. Light Flux Sensors

The SoM includes two Texas Instruments OPT3001 light flux sensors, one on the top side and one on the bottom side of the board. The sensors are connected to the microcontroller over the I^2C bus (`I2C0`), at addresses `0x44` (top) and `0x45` (bottom). It measures light with a spectral response very closely matched to the human eye, and with very good infrared rejection. The relevant performance metrics of the sensors is provided in Table 12.

Table 12: Capabilities of the TI OPT3001 Light Flux Sensor

Parameters	Condition	Min.	Typical	Max.	Unit
Operating temperature		−40		+85	°C
Peak irradiance spectral responsivity			550		nm
Resolution			0.01		$\frac{\mathrm{lux}}{\mathrm{LSB}}$
Full-scale illuminance			83865.6		lux
Dark noise			0	0.03	lux
Half-power angle			47		°

8. Application

8.1. Board Stacking

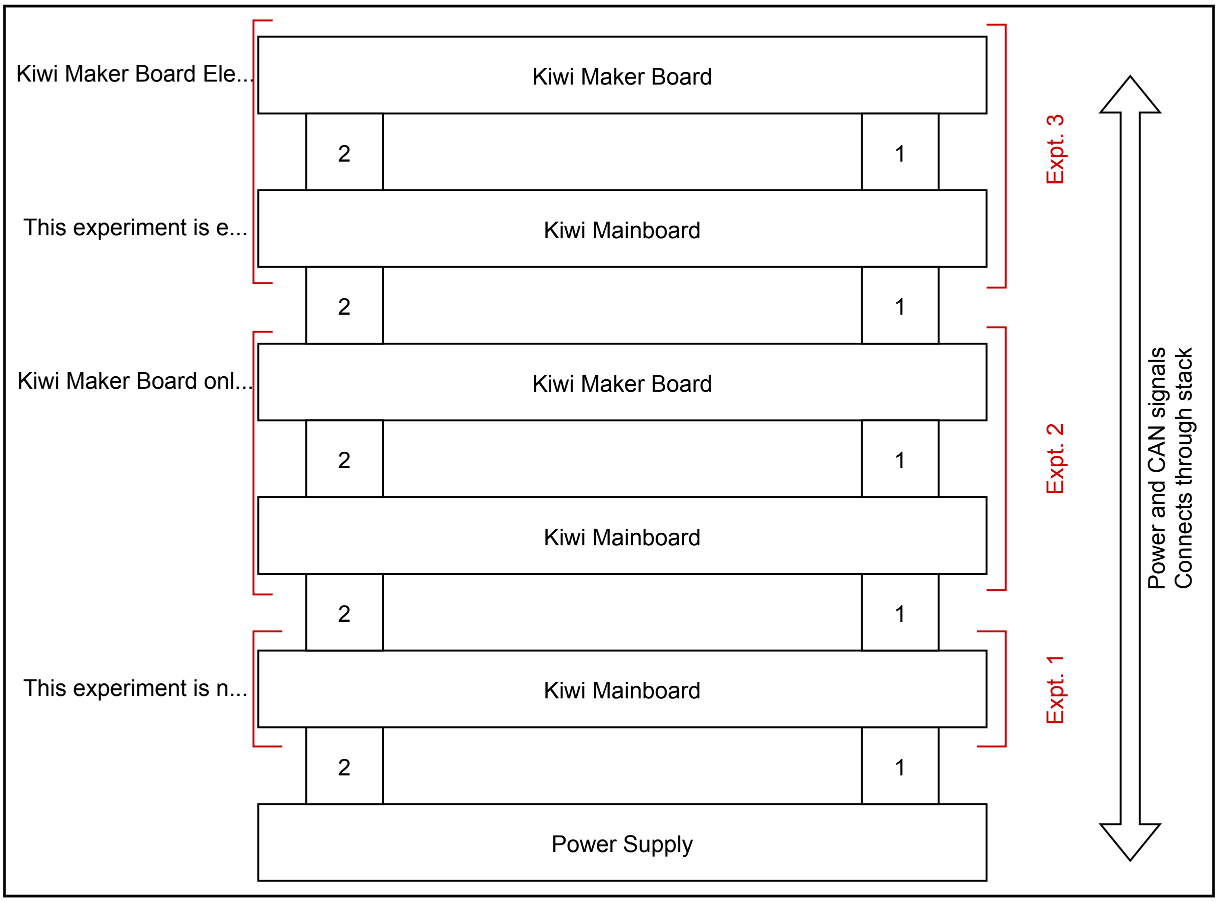

Figure 4: Stacking Kiwi Mainboard and Kiwi Maker Board to combine experiments.

> ⚠️ **Warning**
>
> Use the GPIOs on Stack Connector 2 ONLY IF the signals are connected to an experiment on a corresponding Kiwi Maker Board. Failure to follow this WILL cause interference with other boards on the stack.
>
> Do not force the module onto the stack. Wiggling the module or applying too much pressure may damage it. If the module does not readily press into place, remove it, check for debris and alignment, and try again.
>
> Mating cycle should not exceed 50.

9. Indexing

9.1. Figures

9.2. Tables

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